

28. Find the Boundary of the Image Using Convolution Kernel

Aim

To find the **boundary or edges of an image using a convolution kernel** in OpenCV.

Procedure

1. Read the input image.
2. Convert the image into grayscale.
3. Define a convolution kernel for boundary detection.
4. Apply the kernel to the image using `cv2.filter2D()`.
5. Convert the result into a suitable display range.
6. Display the original and boundary images.
7. Save the boundary-detected image.

Convolution Kernel

A simple boundary detection kernel is:

This kernel highlights pixels that are different from their surrounding pixels, thereby detecting boundaries.

Program

```
import cv2

import numpy as np

# Read input image

img = cv2.imread("input.jpg")

# Check image

if img is None:
```

```
print("Error: Image not found")

exit()

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Boundary detection kernel

kernel = np.array([

    [-1, -1, -1],

    [-1, 8, -1],

    [-1, -1, -1]

], dtype=np.float32)

# Apply convolution

boundary = cv2.filter2D(gray, -1, kernel)

# Convert negative values and normalize

boundary = cv2.convertScaleAbs(boundary)

# Display images

cv2.imshow("Original Image", gray)

cv2.imshow("Boundary Image", boundary)

# Save output
```

```
cv2.imwrite("boundary_image.jpg", boundary)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output



Result

The **boundary of the given image was successfully detected using a convolution kernel.**

The edges and boundaries of objects in the image were highlighted in the output image.